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Multi-Objective Optimization of Analog Circuits Using Reinforcement Learning

Hakan Taşkıran, Enes Sağlıcan, and Engin Afacan
λ-Laboratory, Dept. of Electronics Engineering Gebze Technical University, Kocaeli, Türkiye

***Abstract*—Analog circuit design requires balancing conflicting performance metrics, making an accurate Pareto-Optimal Front (PoF) essential for trade-off analysis. Meta-heuristic algorithms are commonly employed to explore the PoF; however, the computational complexity involved in PoF exploration and the tendency to converge on suboptimal designs remain significant challenges that have yet to be fully addressed. This work introduces Multi-Objective Deep Deterministic Policy Gradient (MODDPG) for efficient PoF extraction in analog circuit design. By combining Reinforcement Learning (RL) with multi-objective optimization, MODDPG navigates high-dimensional design spaces more effectively than the conventional methods. It leverages a continuous action space and a tailored reward function to iteratively refine the design parameters. To validate the performance of the proposed approach, several MOO benchmark problems have been successfully solved. Then, experiments on analog circuit benchmarks have been perfromed to demonstrate that MODDPG outperforms traditional methods in key metrics, notably maximizing the Figure of Merit (FoM) and significantly advancing automation in circuit optimization.**

***Index Terms*—Analog, RF, MOO, RL, EDA ML, CAD, ANN.**

I. Introduction

The design of analog circuits presents a complex multi-objective optimization problem, where conflicting performance metrics such as unity-gain-bandwidth and power consumption, should be concurrently optimized [1], [2]. Achieving an optimal balance among the objectives is critical for ensuring circuit performance while adhering to design constraints. Traditionally, PoF extraction has been employed to characterize trade-offs between objectives, offering circuit designers a set of non-dominated solutions. However, conventional optimization techniques, such as gradient-based methods [3] and heuristic algorithms [4], often face challenges in high-dimensional design spaces and require significant computational resources, particularly when paired with circuit simulators. As a result, the development of efficient optimization algorithms for analog sizing problems remains a prominent area of research. In recent years, RL has emerged as a promising approach for addressing complex optimization problems due to its ability to learn optimal policies through interaction with an environment. Among several RL techniques, the Deep Deterministic Policy Gradient (DDPG) algorithm, designed for continuous action spaces, has been successfully adapted for multi-objective problems, leading to the development of MODDPG. MODDPG extends DDPG by incorporating mechanisms to handle multiple objectives, making it well-suited for extracting the PoF in analog circuit design. In this manner, we propose a framework for using MODDPG to automate PoF extraction in analog circuit design in this work. The framework leverages MODDPG's ability to efficiently navigate the design space, using a reward function that balances proximity to the true Pareto front and diversity among solutions. By integrating this RL-based approach with circuit simulators, the method enables direct evaluation of circuit performance metrics, reducing reliance on surrogate models and improving the fidelity of the optimization process. The contributions of this study are threefold:

- Development of a MODDPG-based approach tailored for PoF extraction in analog/RF circuit design.
- Implementation of a custom reward function to promote both convergence to the Pareto front and solution diversity.
- Validation of the proposed method on ZDT benchmark and analog circuits, demonstrating its efficiency and effectiveness compared to traditional multi-objective optimization techniques.

The remainder of this paper is organized as follows: Section II briefly discusses previous work in the field. Section III presents the proposed approach in detail. Section IV provides demonstration examples and synthesis results. Finally, Section V concludes the paper.

II. Previous Work

Design automation systems are essential to improve efficiency and address the increasing demand for high-quality integrated circuits despite a shortage of trained designers. Research focuses on identifying automation solutions for nonlinear analog/RF circuit designs and algorithms that replicate the expertise of human designers. Over the last two decades, metaheuristic algorithms have been prominent in this domain. According to [2] that classified these algorithms, these include Evolutionary Computation (EC), which follows Darwin's theory of evolution, and Swarm Intelligence (SI), inspired by swarm communication. EC algorithms like Genetic Algorithms, Evolutionary Strategies, Genetic Programming, and Differential Evolution are widely used, alongside SI algorithms such as Particle Swarm Optimization Ant Colony Optimization, and Artificial Bee Colony Optimization.

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Applications of these methods have addressed circuit sizing problems effectively. Integrated circuit optimization methods are categorized into single-objective and multi-objective approaches. Single-objective optimization minimizes a unified cost function. Multi-objective optimization, by contrast, focuses on generating Pareto fronts for conflicting objectives. Analytical methods, like those by [5], gradient-based approaches, and genetic algorithms such as NSGA-II, have proven effective. Hybrid techniques combining Gaussian Processes and Bayesian frameworks [6] or genetic algorithms with surrogate models [7] have further enhanced optimization efficiency. , For example, [8] developed a two-phase method alternating between Bayesian optimization and Pareto pre-selection. More recently, Machine Learning (ML) has gained traction, questioning the effectiveness of traditional metaheuristic methods. Related examples can be found in [9], which explores ML techniques in detail. Support Vector Machines (SVM) and Artificial Neural Networks (ANN) have been used to model analog circuit functions. For example, SVM has been applied to analog circuit modeling and validated through SPICE simulations, while ANN-based approaches have been utilized for modeling power consumption and predicting performance empirically. ANN techniques have also been successfully employed in performance modeling, achieving design goals despite inconsistencies between simulated and desired performance. RL, a rapidly evolving approach, shows the potential to outperform traditional algorithms. Examples include RL-based methods for optimizing analog circuit targets and frameworks such as AutoCkt for System-on-Chip designs [10]. Multi-Objective Reinforcement Learning (MORL) has shown promise in analog circuit optimization. [11] proposed a MORL-based method to optimize Pareto fronts, reducing SPICE simulations and improving efficiency.

However, enhancements in quality metrics and Pareto point density are needed. This study aims to develop a sample-efficient MORL algorithm to automate design, minimize human intervention, and enable reusable Pareto agents. Expected outcomes include fewer SPICE simulations, faster design cycles, competitive performance with genetic algorithms, and industrial applicability for analog ICs across various technology nodes. It is important to note that all syntheses in this study were performed using the UMC 130nm CMOS process.

III. The Proposed Approach

The proposed methodology employs the MODDPG in Alg. 1 to tackle multi-objective circuit design optimization, addressing conflicting objectives such as gain, bandwidth, and power consumption without scalarized rewards. Instead, vectorized rewards retain trade-offs, enabling more precise optimization. The ***state*** represents circuit parameters, while ***actions*** correspond to incremental adjustments suggested by the actor network. The ***reward***, a multi-objective performance vector, prevents information loss from scalarization. The ***agent*** comprises an actor network for action generation and two critic networks for objective evaluation, interacting with a circuit simulation environment that updates the state and computes vectorized rewards. A replay buffer stores experiences to stabilize learning and enhance exploration. The algorithm ranks actions via Pareto dominance, prioritizing non-dominated solutions while ensuring diversity through a crowding distance metric. Prioritized replay sampling guides exploration toward Pareto-optimal solutions. Soft updates of target networks ensure stability and efficiency. By leveraging vectorized rewards and multi-objective ranking, this approach preserves problem complexity, facilitating the discovery of diverse, optimal circuit designs. Also, the general structure of the proposed approach is depicted in Fig. 1.

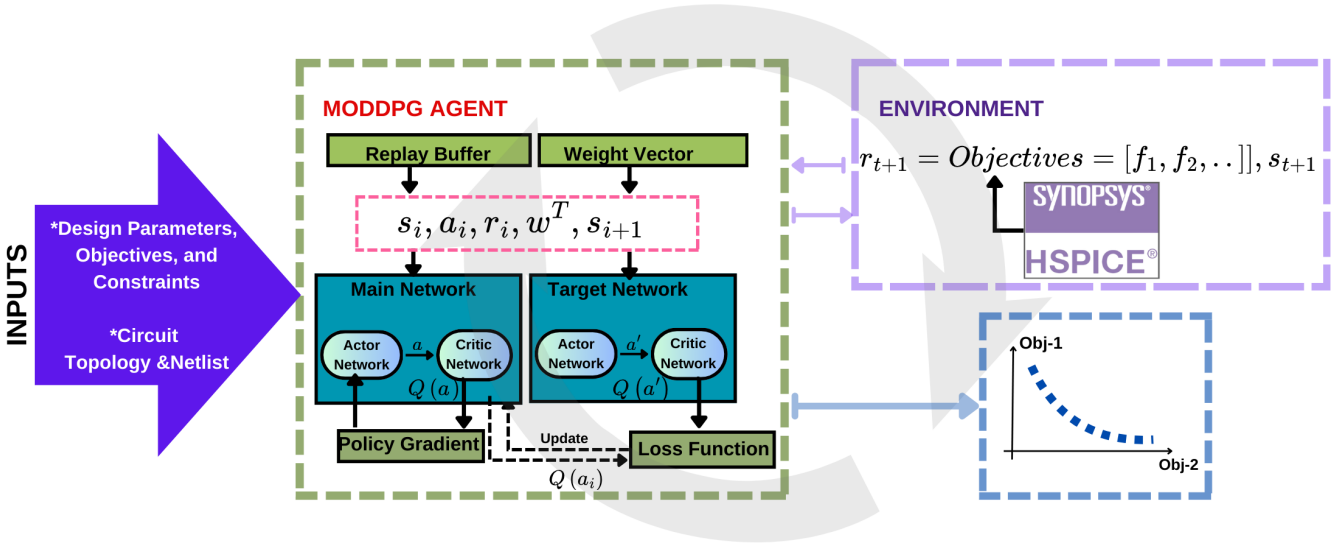


Figure 1: The general framework of the proposed approach.

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Algorithm 1 MODDPG for Pareto Front Generation

Initialize: Actor network $\mu(s|\theta^{\mu})$, Critic networks $Q_1(s,a|\phi_1)$, $Q_2(s,a|\phi_2)$
Initialize: Target networks $\mu'(s|\theta^{\mu'})$, $Q_1'(s,a|\phi_1')$, $Q_2'(s,a|\phi_2')$
Initialize: Replay buffer $\mathscr{R}$ and hyperparameters α_μ, α_Q, γ, τ, B, σ
Set: Multi-objective weights $\mathbf{w} = [\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_k]$
for Each episode **do**
  Reset environment and observe initial state s_0
  while Episode not done **do**
    Select action $a_t = \mu(s_t|\theta^{\mu}) + \mathscr{N}_t$, where $\mathscr{N}_t \sim \mathscr{N}(0,\sigma)$
    Execute action a_t and observe reward $\mathbf{r}_t = [r_{1,t}, \ldots, r_{k,t}]$ and next state s_{t+1}
    Store transition $(s_t, a_t, \mathbf{r}_t, s_{t+1})$ in $\mathscr{R}$
  end while
  for Each training step **do**
    Sample mini-batch $\{(s_i, a_i, \mathbf{r}_i, s_{i+1})\}_{i=1}^{B}$ from $\mathscr{R}$
    Compute target value for critic update:

$$y_i = \mathbf{r}_i + \gamma \min_{j=1,2} Q_j'(s_{i+1}, \mu'(s_{i+1}|\theta^{\mu'})|\mathbf{w})$$

    Compute loss for critic networks:

$$\mathscr{L}_{Q_j} = \frac{1}{B}\sum_{i=1}^{B} \|y_i - Q_j(s_i, a_i|\phi_j)\|^2 \quad \text{for } j \in \{1,2\}$$

    Update critic parameters:

$$\phi_j \leftarrow \phi_j - \alpha_Q \nabla_{\phi_j} \mathscr{L}_{Q_j} \quad \text{for } j \in \{1,2\}$$

    Compute loss for actor network using deterministic policy gradient:

$$\mathscr{L}_{\mu} = -\frac{1}{B}\sum_{i=1}^{B} Q_1(s_i, \mu(s_i|\theta^{\mu})|\phi_1)$$

    Update actor parameters:

$$\theta^{\mu} \leftarrow \theta^{\mu} - \alpha_\mu \nabla_{\theta^{\mu}} \mathscr{L}_{\mu}$$

    Apply Pareto-based prioritization:
    Compute objective values $\mathbf{f}(a) = [f_1(a), \ldots, f_k(a)]$ for all actions $a \in \mathscr{R}$
    Perform non-dominated sorting to assign Pareto ranks
    Compute crowding distances for each action within the same rank
    Sample actions based on Pareto-optimal prioritization
    Soft update target networks:

$$\theta^{\mu'} \leftarrow \tau\theta^{\mu} + (1-\tau)\theta^{\mu'}$$

$$\phi_j' \leftarrow \tau\phi_j + (1-\tau)\phi_j \quad \text{for } j \in \{1,2\}$$

  end for
end for
Generate Pareto front from replay buffer $\mathscr{R}$ using non-dominated sorting

IV. Case Studies

To validate the MODDPG approach, we first evaluated its performance on standard ZDT benchmark problems. The optimization results for these benchmarks are presented in Fig. 2.

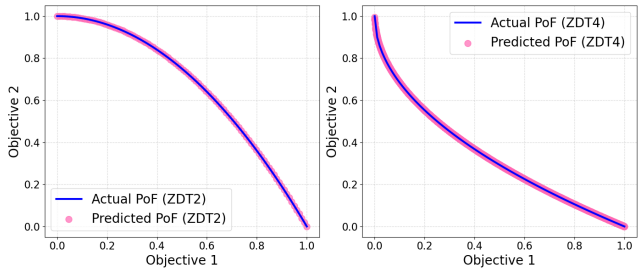


Figure 2: Verification of the MODDPG on the ZDT problems.

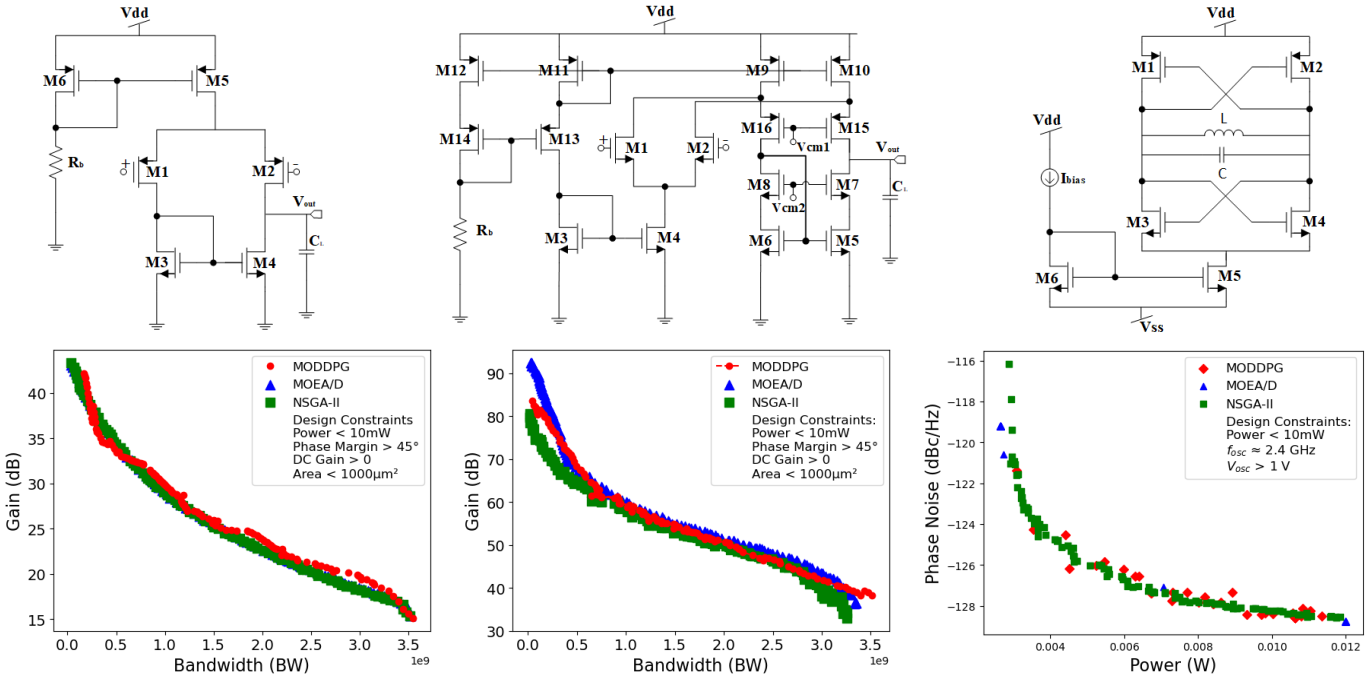
As seen from the results, the results of the developed approach perfectly align with the actual Pareto-optimal points. After verifying its efficacy on theoretical problems, we applied MODDPG to three analog circuit design tasks: Active-Loaded Differential Amplifier, Folded Cascode Amplifier, and CMOS LC Oscillator, as shown in Fig. 3. For the amplifiers, the objective was to optimize bandwidth and gain while adhering to design constraints and parameter boundaries. In the CMOS LC Oscillator design, where parasitic-aware equivalent circuits are used for passive devices as in [12], we set the oscillation frequency to 2.4 GHz while imposing constraints on phase noise, power consumption, and oscillation amplitude to achieve efficiency. The amplitude constraint was particularly crucial for sustaining oscillation, emphasizing the voltage-limited mode to reach an optimal balance between power consumption and phase noise. All the design objectives and the constraints are provided on the POF results given in Fig. 3. Finally, the MODDPG algorithm was employed to extract the PoF, ensuring that all generated solutions satisfied the predefined design constraints and optimization objectives. To evaluate the performance of the proposed MODDPG framework, it has been compared against well-known evolutionary computation (EC) algorithms, specifically MOEA/D and NSGA-II. For this purpose, all benchmark circuits were also synthesized using these EC algorithms adopted from [13]. Performance is assessed using the Number of Pareto-optimal solutions (PoS) and simulation (NoS), Hypervolume (HV), Generational Distance (GD), Inverted Generational Distance (IGD), and Spacing (S) as shown in Table I. MOEA/D achieves the highest HV for the differential amplifier and the folded-cascode amplifier, while NSGA-II excels in the Oscillator circuit. The MODDPG demonstrates strong convergence, achieving the lowest GD and IGD values for both the folded-cascode amplifier and the oscillator. Moreover, MODDPG consistently produces well-distributed solutions, achieving the lowest spacing values across all circuits. In terms of area, MODDPG attains the highest value for the differential amplifier and shares the best value with MOEA/D for the folded-cascode amplifier. On the other hand, NSGA-II performs the best for the oscillator. To evaluate these performance metrics holistically, a Figure of Merit (FoM) is defined, as shown in Equation (1). As illustrated in the last row of Table I, NSGA-II performs best for the differential amplifier, which is a relatively simpler problem, while the proposed approach outperforms the other algorithms when the problem becomes more complex.

$$FoM = \frac{HV \cdot \text{PoS}}{GD \cdot IGD \cdot S \cdot \text{NoS}} \tag{1}$$

V. Conclusion

The proposed MODDPG framework offers a promising solution to the challenges of efficient PoF extraction in analog circuit design. By integrating RL with multi-objective optimization, MODDPG effectively navigates complex design spaces and overcomes the limitations of conventional methods. The experimental results demonstrated that MODDPG outperforms the other well-known algorithms especially for the complicated design problems, which reveals the

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(a) Differential Amplifier (b) Folded Cascode Amplifier (c) Cross-Coupled LC Oscillator

Figure 3: Optimization results of the proposed RL-MODDPG, the NSGA-II, and the MOEA/D.

Table I: Comparison of Differential Amplifier, Folded Amplifier, and Oscillator with MOEA/D, NSGA-II, and MODDPG

	Differential Amplifier			Folded Amplifier			Oscillator		
Metric	MOEA/D	NSGA-II	MODDPG	MOEA/D	NSGA-II	MODDPG	MOEA/D	NSGA-II	MODDPG
HV	**0.63**	0.62	0.58	**0.58**	0.40	0.51	0.07	**0.17**	0.15
GD	0.06	0.06	0.07	0.04	0.02	**0.01**	0.03	0.03	**0.01**
IGD	**0.05**	**0.05**	0.07	0.02	0.02	**0.01**	0.03	0.03	**0.01**
S	0.007	**0.006**	**0.006**	0.007	0.006	**0.005**	0.09	0.07	**0.05**
PoS	101	100	**132**	**150**	**150**	**150**	4	**101**	78
NoS	10e3	10e3	**10e3**	60e3	60e3	**15e3**	40e3	40e3	**20e3**
FoM	3.03	**3.44**	2.6	2.59	4.16	**102**	0.086	6.81	**117**

*HV: Hypervolume, GD: Generational Distance, IGD: Inverted Generational Distance, S: Spacing, PoS: Pareto optimal Solutions, NoS: Number of Simulation.

potential of MODDPG to transform analog circuit design by providing more accurate and well-distributed POFs.

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